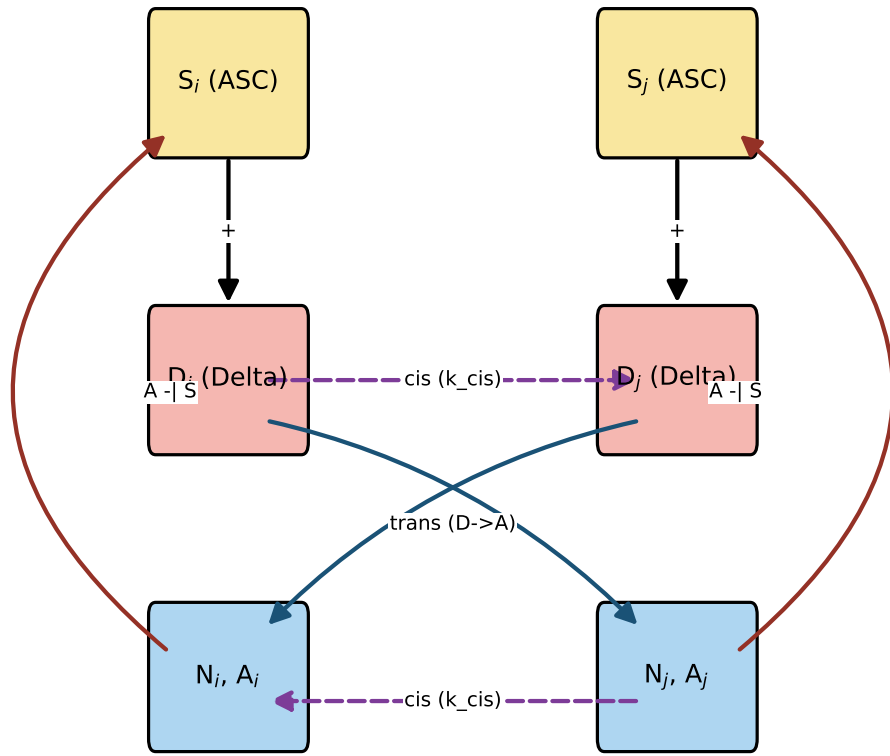
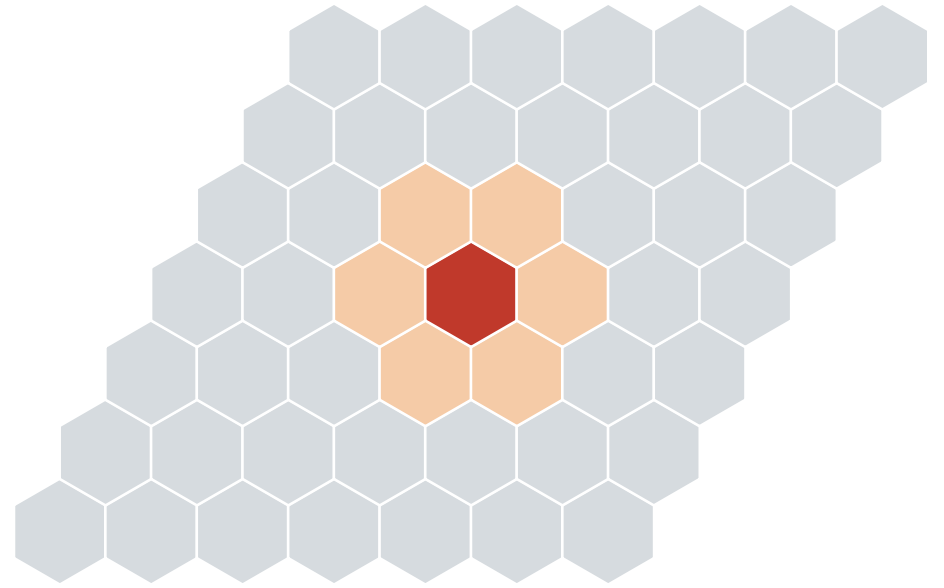


Figure 1 -- Biological circuit and tissue geometry

A. Feedback circuit (cell i, neighbour j)

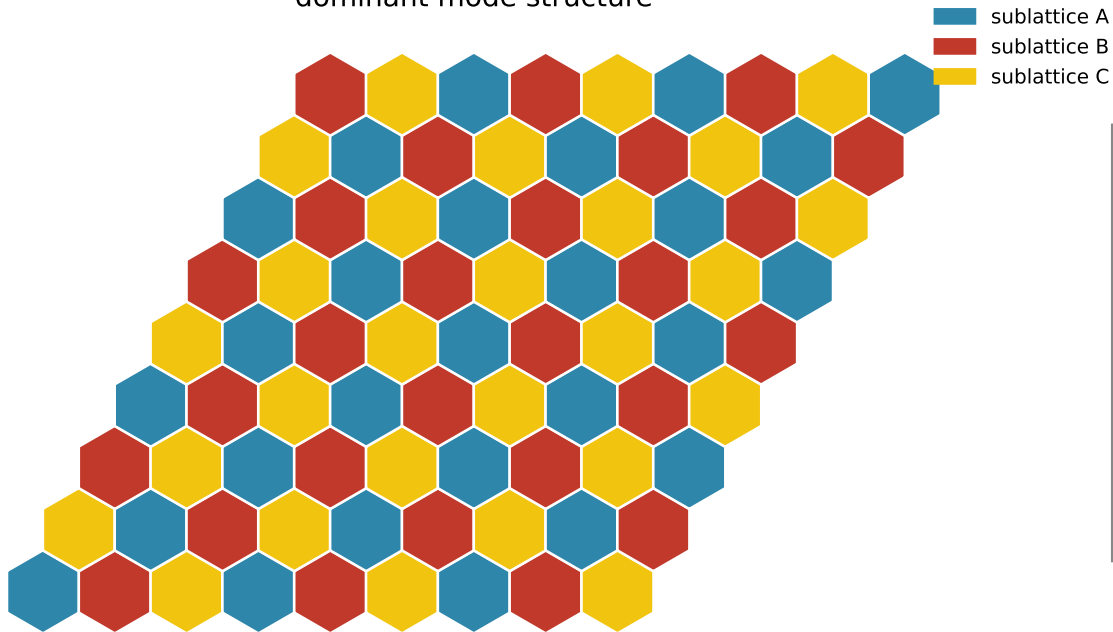


B. Hexagonal tiling: one cell's six contact neighbours



D. Governing equations (exact, model.equations)
+ block preview

C. Three-sublattice (A/B/C)
dominant mode structure



$$\begin{aligned}\dot{D} &= \lambda_D h_D(S) - d_D D - f_D D s_N - k_{cis} N D \\ \dot{N} &= \lambda_N - d_N N - f_N N s_D - k_{cis} N D \\ \dot{A} &= -d_A A + f_N N s_D \\ \dot{S} &= -d_S S + \lambda_S h_S(A)\end{aligned}$$

$$s_D = \sum_{j \sim i} D_j, \quad s_N = \sum_{j \sim i} N_j$$

Small-signal block preview:

$$\dot{z} = A_\mu z + bS, \quad \dot{S} = c^\top z + a_S S$$